

GRAHAM, TX, USA

WMO: 720318

Lat: 33.110N

Lon: 98.555W

Elev: 342

StdP: 97.28

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53965

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.3	(c) -5.1	(d) -16.2	(e) 0.9	(f) 0.8	(g) -12.5	(h) 1.3	(i) 2.8	(j) 11.1	(k) 7.7	(l) 9.8	(m) 6.7	(n) 2.1	(o) 350	(p) 0.437

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	39.1	22.1	37.7	21.9	36.4	22.2	25.2	32.4	24.7	31.8	24.2	31.3	3.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.7	19.3	27.3	22.9	18.4	26.9	22.4	17.9	26.8	79.1	32.4	76.6	31.8	74.6	31.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.2	(b) 8.0	(c) 7.1	(e) -11.8	(f) 41.8	(g) 2.4	(h) 2.4	(i) -13.5	(j) 43.5	(k) -14.9	(l) 44.9	(m) -16.2	(n) 46.2	(o) -17.9	(p) 47.9		
(a) 9.2	(b) 8.0	(c) 7.1	(e) -12.2	(f) 26.8	(g) 2.2	(h) 3.1	(i) -13.7	(j) 29.0	(k) -15.0	(l) 30.9	(m) -16.2	(n) 32.6	(o) -17.8	(p) 34.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.4	6.7	9.4	14.2	18.3	22.4	27.4	29.2	29.0	25.3	18.6	12.6	7.4
	DBStd	9.32	5.57	6.71	5.79	4.81	4.33	2.78	2.48	2.83	3.64	5.27	5.46	5.60
	HDD10.0	409	133	87	26	4	0	0	0	0	0	5	36	119
	HDD18.3	1433	362	258	152	58	16	0	0	0	2	62	182	341
	CDD10.0	3482	30	68	156	253	386	523	594	590	459	272	114	38
	CDD18.3	1464	1	7	23	58	142	273	336	332	211	71	11	2
	CDH23.3	17619	21	96	259	646	1506	3266	4360	4238	2337	753	113	24
	CDH26.7	9161	2	24	79	231	658	1731	2519	2446	1167	273	28	3
Wind	WSAvg	2.8	2.8	3.1	3.5	3.7	3.1	2.8	2.4	2.2	2.3	2.6	2.7	2.7
Precipitation	PrecAvg	802	36	47	64	71	96	103	71	59	71	79	53	41
	PrecMax	1120	83	135	153	208	206	266	164	205	155	194	213	152
	PrecMin	512	1	3	2	2	20	11	3	0	8	9	0	0
	PrecStd	155	25	40	41	52	46	60	49	47	44	52	56	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.0	29.9	32.1	35.1	37.3	39.5	41.4	41.0	38.0	33.7	30.1	26.5
		MCWB	13.4	15.2	16.8	18.2	22.4	22.0	21.7	21.6	21.8	21.3	17.8	13.2
	2%	DB	22.4	26.3	28.9	31.8	34.8	37.4	39.0	39.1	36.2	32.1	26.3	22.5
		MCWB	11.9	15.2	16.4	18.0	20.8	21.5	21.9	22.3	21.8	20.8	16.2	12.6
	5%	DB	19.5	23.2	26.3	29.3	32.5	35.9	37.5	37.8	34.6	29.9	23.8	19.8
		MCWB	11.5	14.2	15.8	18.1	20.8	22.1	21.8	22.1	21.9	19.7	15.8	12.3
	10%	DB	17.0	20.4	23.7	27.2	30.6	34.1	36.1	36.3	32.8	27.7	21.7	17.1
		MCWB	10.3	13.0	14.8	17.3	20.5	22.8	22.6	22.6	21.5	18.5	14.5	10.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.3	19.1	20.1	22.4	24.9	26.4	26.1	25.5	25.1	24.2	20.7	18.5
		MCDB	20.3	24.0	26.7	28.1	31.0	33.4	33.4	32.5	31.3	29.4	24.0	21.3
	2%	WB	14.6	17.5	18.8	21.1	23.7	25.2	25.2	24.8	24.2	22.9	18.6	15.6
		MCDB	19.0	22.2	24.9	26.6	29.9	31.6	32.9	32.6	30.6	28.6	21.8	19.2
	5%	WB	12.4	15.9	17.7	20.3	22.9	24.4	24.6	24.3	23.7	21.8	17.5	13.5
		MCDB	18.3	21.3	22.8	25.6	29.0	30.9	32.2	32.1	29.9	27.3	21.3	17.6
	10%	WB	10.4	14.2	16.5	19.4	22.1	23.8	24.0	23.9	23.0	20.7	16.2	11.3
		MCDB	16.1	19.2	21.8	24.5	28.0	30.4	31.5	31.6	29.1	25.6	20.8	16.5
Mean Daily Temperature Range	5% DB	MDBR	13.7	13.6	13.9	14.1	12.7	12.2	12.9	13.0	12.9	13.9	13.9	13.0
		MCDBR	18.2	17.5	16.5	16.8	15.5	14.0	14.7	14.6	14.5	15.1	15.9	17.3
		MCWBR	10.1	8.9	7.6	7.2	5.4	3.7	3.5	3.5	4.1	6.4	8.1	9.3
	5% WB	MCDBR	15.4	15.3	12.9	12.7	11.8	11.7	12.2	12.7	11.6	12.3	12.1	13.3
		MCWBR	9.9	9.0	6.6	7.0	5.2	4.5	3.8	3.6	4.0	6.0	8.2	8.6
Clear-Sky Solar Irradiance	taub	0.297	0.307	0.337	0.373	0.410	0.420	0.424	0.416	0.386	0.350	0.320	0.301	
	taud	2.533	2.515	2.445	2.366	2.331	2.335	2.371	2.355	2.401	2.487	2.506	2.525	
	Ebn at Noon	928	952	941	915	880	866	860	864	878	889	891	901	
	Edh at Noon	83	93	108	122	128	127	122	122	112	95	84	79	
All-Sky Solar Radiation	RadAvg	3.05	3.64	4.82	5.83	6.21	6.90	6.90	6.28	5.25	4.13	3.21	2.66	
	RadStd	0.32	0.55	0.32	0.47	0.63	0.59	0.44	0.39	0.44	0.61	0.41	0.32	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page